

ERDŐS 123⁺: DISTINCT $2^k 3^\ell 5^m$ SUMMANDS WITHIN A FACTOR $1 + \varepsilon$

PRINCIPIA MATH

ABSTRACT. We record a one-line corollary of the representation theorem behind our solution of Erdős Problem 123: for $\{2^k 3^\ell 5^m\}$ and every $\varepsilon > 0$, every sufficiently large integer is a sum of distinct such numbers whose largest and smallest terms differ by a factor less than $1 + \varepsilon$. This settles the stronger conjecture Erdős recorded in 1992 ([1]; the comment to erdosproblems.com/123), by applying that theorem with $\rho = 1 + \varepsilon$.

THE THEOREM WE PROVED

Let $a, b, c \geq 2$ be pairwise coprime and $\mathcal{S} = \{a^k b^\ell c^m : k, \ell, m \in \mathbb{Z}_{\geq 0}\}$; write $d = \min(a, b, c)$. Our solution of Erdős 123 rests on the following theorem, formalized and machine-checked in Lean 4 (Theorems 1.1 and 1.2 of *Antichain Subset Sums in Rank-Three Multiplicative Semigroups*; for rational ρ it comes from a uniform local central limit theorem for the Bernoulli subset sums of $\mathcal{S} \cap [x, \rho x)$).

Theorem 1. *For every real ρ with $1 < \rho < d$, every sufficiently large integer N can be written as*

$$N = \sum_{s \in A} s, \quad A \subset \mathcal{S} \cap [x, \rho x)$$

for some positive integer x . Since any two elements of $[x, \rho x)$ have ratio $< \rho < d$, no element of A divides another; thus the terms of A are distinct and no one of them divides another.

THE STRENGTHENING, AND HOW IT FOLLOWS

For $(a, b, c) = (2, 3, 5)$, Erdős asked for more than distinctness and non-divisibility: he conjectured that all the summands can be taken from a single interval $[x, \rho x)$ with ρ as close to 1 as one wishes.

Corollary 1 (Erdős 123⁺, [1], 1992). *For every $\varepsilon > 0$, every sufficiently large integer n can be written as a sum of distinct integers $b_1 < \cdots < b_t$ of the form $2^k 3^\ell 5^m$ with*

$$b_t < (1 + \varepsilon) b_1.$$

Proof. Here $d = \min(2, 3, 5) = 2$. Given $\varepsilon > 0$, set

$$\rho = 1 + \min(\varepsilon, \tfrac{1}{2}) \in (1, 2),$$

so that $1 < \rho < d$ and Theorem 1 applies. For all large n it gives a set $A \subset \mathcal{S} \cap [x, \rho x)$ with $\sum_{s \in A} s = n$; list its distinct elements as $b_1 < \cdots < b_t$. Every b_i lies in $[x, \rho x)$, so $b_1 \geq x$ and

$$b_t < \rho x \leq (1 + \varepsilon) x \leq (1 + \varepsilon) b_1.$$

The terms are distinct and pairwise non-dividing by the last sentence of Theorem 1. (When $\varepsilon \geq 1$ the choice $\rho = \frac{3}{2}$ already gives the claim.) $\square$

The only freedom used is the choice of ρ : the two endpoints of the interval $[x, \rho x)$ in Theorem 1 are in the ratio ρ , which we take to be $1 + \varepsilon$. The same argument gives the corresponding statement for every pairwise-coprime triple a, b, c , since a ratio $\rho = 1 + \varepsilon < d$ is available whenever $\varepsilon < d - 1$.

REFERENCES

- [1] P. Erdős, conjecture recorded in 1992; see the comments to Erdős Problem 123 at <https://www.erdosproblems.com/123> (reference [Er92b]).